CURRICULUM VITAE

Yiwen Hu

EDUCATION

Ph.D.	2025	Michigan State University (MSU), Computer Science
B.E.	2018	Zhejiang University (ZJU), Digital Media Technology (College of Computer Science & Technology)

Experience in Higher Education

2025-present	University of Maryland, Baltimore County	Assistant Professor, Department of Computer Science and Electrical Engineering
2018-2025	Michigan State University	Graduate Research Assistant, Department of Computer Science and Engineering
2024 (Fall)	Michigan State University	Guest Lecturer, Department of Computer Science and Engineering
2020-2022	Michigan State University	Graduate Teaching Assistant, Department of Computer Science and Engineering

Honors Received

2025	ACM GetMobile Research Highlights
2025	Recognition for Outstanding Research, MSU
2025	Android High-Severity Security Issues Report, Google
2024	Dissertation Completion Fellowship, MSU
2024	ACM SIGMOBILE Research Highlights (Primary Author)
2024	Graduate School Travel Grant, MSU
2023	ACM GetMobile Research Highlights (Primary Author)
2023	Contribution to Standards Acknowledgment, GSMA
2023	AT&T Security Reward (Primary Author)
2023	Graduate Fellowship, MSU
2022	ACM MobiCom Best Community Paper Award Runner-Up (Primary Author)
2022	ACM MobiCom Student Travel Grant
2022	Carl V.Page Graduate Memorial Fellowship

Research Support

- Startup, source: UMBC.

Ph.D. Students

- Chuanzhi Zhu, Fall 2026–Spring 2031 (exp.), UMBC, Chair
- Haitian Yan, Fall 2026–Spring 2031 (exp.), UMBC, Chair

Master's Students

- Nicole Lippitt, UMBC, Spring 2026–Present, Committee Member
- Demarko Fulcher, UMBC, Fall 2025–Present, Research Project Mentor
- Dharani Nadendla, UMBC, Fall 2025–Present, Research Project Mentor

Undergraduate Students

- Shayne Hewitt, UMBC, Fall 2025–Present, Research Project Mentor
- Bella Goltser, UMBC, Fall 2025–Present, Research Project Mentor
- Dipa Myae, UMBC, Fall 2025–Present, Research Project Mentor
- Olaoluwa Ogunsanya, UMBC, Fall 2025–Present, Research Project Mentor

PUBLICATIONS, PRESENTATIONS, AND CREATIVE ACHIEVEMENTS

Publications

Peer-Reviewed Conference Proceedings

[C8] Min-Yue Chen, Yiwen Hu, Yu-An Chen, Chi-Yu Li, Tian Xie, Guan-Hua Tu, Insecurity of Lost/Stolen Phone Reporting Services: Vulnerabilities, Attacks, and Countermeasures, *In Proceedings of the 24th International Conference on Mobile Systems, Applications, and Services*, 2026 Conditionally Accepted, ACM MobiCom 2026

[C7] Yiwen Hu, Min-Yue Chen, Haitian Yan, Chuan-Yi Cheng, Guan-Hua Tu, Chi-Yu Li, Tian Xie, Chunyi Peng, Li Xiao, and Jiliang Tang. Uncovering problematic designs hindering ubiquitous cellular emergency services access. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*, pages 1455–1469, 2024
https://dl.acm.org/doi/abs/10.1145/3636534.3690704 ACM MobiCom 2024

[C6] Jingwen Shi, Sihan Wang, Min-Yue Chen, Guan-Hua Tu, Tian Xie, Man-Hsin Chen, Yiwen Hu, Chi-Yu Li, and Chunyi Peng. Ims is not that secure on your 5g/4g phones. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*, pages 513–527, 2024
https://dl.acm.org/doi/abs/10.1145/3636534.3649377 ACM MobiCom 2024

[C5] Jingwen Shi, Tian Xie, Guan-Hua Tu, Chunyi Peng, Chi-Yu Li, Andrew Hou, Sihan Wang, Yiwen Hu, Xinyu Lei, Min-Yue Chen, et al. When good turns evil: Encrypted 5g/4g voice calls can leak your identities. In *2023 IEEE Conference on Communications and Network Security (CNS)*, pages 1–9. IEEE, 2023
https://ieeexplore.ieee.org/abstract/document/10288900...9ZSQ IEEE CNS 2023

[C4] Xinyang Li, Xinyu Lei, Yantao Li, Huafeng Qin, and Yiwen Hu. Towards inference of original graph data information from graph embeddings. In *2023 International Joint Conference on Neural Networks (IJCNN)*, pages 1–10. IEEE, 2023
https://ieeexplore.ieee.org/abstract/document/10191065...o0I4 IEEE IJCNN 2023

[C3] Yiwen Hu, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie, Li Xiao, Chunyi Peng, Zhaowei Tan, et al. Uncovering insecure designs of cellular emergency services (911). In *Proceedings of the 28th Annual International Conference on Mobile Computing And Networking*, pages 703–715, 2022
https://dl.acm.org/doi/abs/10.1145/3495243.3560534 ACM MobiCom 2022

[C2] Sihan Wang, Guan-Hua Tu, Xinyu Lei, Tian Xie, Chi-Yu Li, Po-Yi Chou, Fucheng Hsieh, Yiwen Hu, Li Xiao, and Chunyi Peng. Insecurity of operational cellular iot service: New vulnerabilities, attacks, and countermeasures. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*, pages 437–450, 2021
https://dl.acm.org/doi/abs/10.1145/3447993.3483239 ACM MobiCom 2021

[C1] Yiwen Hu, Sihan Wang, Guan-Hua Tu, Li Xiao, Tian Xie, Xinyu Lei, and Chi-Yu Li. Security threats from bitcoin wallet smartphone applications: Vulnerabilities, attacks, and countermeasures. In *Proceedings of the Eleventh ACM Conference on Data and Application Security and Privacy*, pages 89–100, 2021
https://dl.acm.org/doi/abs/10.1145/3447993.3483239 ACM CODASPY 2021

Peer-Reviewed Journals

[J4] Yantao Li, Xinyang Li, Xinyu Lei, Huafeng Qin, Yiwen Hu, and Gang Zhou. On the inference of original graph information from graph embeddings. *ACM Transactions on Sensor Networks*, 20(5):1–28, 2024
https://dl.acm.org/doi/full/10.1145/3688846 ACM TOSN 2024

[J3] Min-Yue Chen, Yiwen Hu* (*:Co-Primary), Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie, Ren-Chieh Hsu, Li Xiao, Chunyi Peng, et al. Taming the insecurity of cellular emergency services (9–1-1): From vulnerabilities to secure designs. *IEEE/ACM Transactions on Networking*, 32(4):3076–3091, 2024
https://ieeexplore.ieee.org/abstract/document/10479537...FqDs IEEE/ACM TON 2024

[J2] Sihan Wang, Tian Xie, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Xinyu Lei, Po-Yi Chou, Fucheng Hsieh, Yiwen Hu, Li Xiao, et al. Dissecting operational cellular iot service security: Attacks and defenses. *IEEE/ACM Transactions on Networking*, 32(2):1229–1244, 2023
https://ieeexplore.ieee.org/abstract/document/10255644...oZuE IEEE/ACM TON 2023

[J1] Xiao Zhang, Griffin Klevering, Xinyu Lei, Yiwen Hu, Li Xiao, and Guan-Hua Tu. The security in optical wireless communication: A survey. *ACM Computing Surveys*, 55(14s):1–36, 2023
https://dl.acm.org/doi/full/10.1145/3594718 ACM CSUR 2023

Invited Articles

[I2] Jingwen Shi, Sihan Wang, Min-Yue Chen, Yiwen Hu, Guan-Hua Tu, Tian Xie, Man-Hsin Chen, Chi-Yu Li, and Chunyi Peng. Ims is not that secure on your 5g/4g phones. *GetMobile: Mobile Computing and Communications*, 29(1):20–24, 2025
https://dl.acm.org/doi/abs/10.1145/3733892.3733899 ACM GetMobile 2025

[I1] Yiwen Hu, Min-Yue Chen, Guan-Hua Tu, Chi-Yu Li, Sihan Wang, Jingwen Shi, Tian Xie, Li Xiao, Chunyi Peng, Zhaowei Tan, et al. Unveiling the insecurity of operational cellular emergency services (911): Vulnerabilities, attacks, and countermeasures. *GetMobile: Mobile Computing and Communications*, 27(1):39–43, 2023
https://dl.acm.org/doi/abs/10.1145/3599184.3599195 ACM GetMobile 2023

Ph.D. Dissertation

[D1] Yiwen Hu. *Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications Over Cellular Networks*. PhD thesis, Michigan State University, 2025 .https://www.proquest.com/openview...gscholar&cbl=18750&diss=y

Peer-Reviewed Conference Work Submitted for Publication (Since Joining UMBC)

[S3] Ke Xie, Xingyi Zhao, Yiwen Hu, Munshi Saifuzzaman, Wen Li, Shuhan Yuan, Tian Xie, and Guan-Hua Tu. Cellsecinspector: Safeguarding cellular networks via automated security analysis on specifications. *arXiv preprint arXiv:2512.24682*, 2025
Submitted to ACM MobiCom 2026, Selected for One-Shot-Revision

[S2] Ziao Jiao, Xinyu Lei, Xu Zhou, Haoyang Chen, Yiwen Hu, Xiao Yang, Transferable Model Parasitism Attack: Exploiting ML Services Without Paying, 2025
Submitted to ACM AsiaCCS 2026, Selected for One-Shot-Revision

[S1] Ke Xie, Xingyi Zhao, Yiwen Hu, Shuhan Yuan, Tian Xie CellularSpecSec-Bench: A Staged Benchmark for Evidence-Grounded Interpretation and Security Reasoning over 3GPP Specifications, 2025
Submitted to ACL 2026, Under Review

Peer-Reviewed Journal Work Submitted for Publication (Since Joining UMBC)

[R2] Jingwen Shi, Sihan Wang, Guan-Hua Tu, Tian Xie, Yiwen Hu, Haitian Yan, Chi-Yu Li, Chunyi Peng, When Mobile Equipment Security Lags Behind Infrastructure: Vulnerabilities, Attacks, and Countermeasures in IMS Services, 2025
Submitted to IEEE TMC 2026, Under Review

[R1] Munshi Saifuzzaman, Ke Xie, Xiao Zhang, Yiwen Hu, Xinyu Lei, Guan-Hua Tu, Sihan Wang, Awen Li, Tian Xie, The Security Vulnerabilities and Countermeasures in End-to-end 5G/4G Cellular Networks: A Surveys, 2026
Submitted to ACM Computing Surveys, 2026, Under Review

Presentations

Peer-Reviewed Conference Presentations

- Yiwen Hu, "Uncovering Problematic Designs Hindering Ubiquitous Cellular Emergency Services Access", ACM MobiCom 2024, Washington, D.C., USA, Nov. 2024
- Yiwen Hu, "Insecurity of operational cellular IoT service: New vulnerabilities, attacks, and countermeasures", ACM MobiCom 2021, New Orleans, LA, USA, Mar. 2022
- Yiwen Hu, "Security threats from bitcoin wallet smartphone applications: Vulnerabilities, attacks, and countermeasures", CODASPY 2021, Taking place virtually, Apr. 2021

Other Professional Presentations

- Yiwen Hu, "Uncovering Problematic Designs Hindering Ubiquitous Cellular Emergency Services Access", Federal Communications Commission, Taking place virtually, Dec. 2025

- Yiwen Hu, "Uncovering Insecure Designs of Cellular Emergency Services (911)", UMBC CDL Talk, UMBC, Sept. 2025
- Yiwen Hu, "Uncovering Insecure Designs of Cellular Emergency Services (911)", Seminar, Michigan State University, East Lansing, MI, May 2025
- Yiwen Hu, "Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks", Colloquium, The University of Iowa, Iowa City, IA, Mar. 2025
- Yiwen Hu, "Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks", Colloquium, The University of Maryland, Baltimore County, Baltimore, MD, Mar. 2025
- Yiwen Hu, "Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks", Colloquium, The University of Nevada Reno, Reno, NV, Mar. 2025
- Yiwen Hu, "Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks", Colloquium, Wayne State University, Detroit, MI, Mar. 2025
- Yiwen Hu, "Powering the Infrastructure for Critical Services: Ensuring Secure and Reliable Emergency Communications over Cellular Networks", Colloquium, The University of Minnesota Twin Cities, Minneapolis, MN, Feb. 2025
- Yiwen Hu, "Uncovering Insecure Designs of Cellular Emergency Services (911)", Third Generation Partnership Project (3GPP, Telecommunications Standards Organizations), Taking place virtually, Oct. 2023

Media Activities

- MSUToday(2025). "MSU researchers identify why 911 calls are delayed, failed, or dropped". March 27.
- ISSSource (2023). "Boosting Security For Cellular Emergency Calls". May 1.
- TheWILS Morning Wake-Up (2023). "The National Science Foundation awarding MSU just over 1 million dollars, what are they using the money on?". April 28.
- Homeland Security News Wire (2023). "Making Emergency Calls More Secure". April 27.
- WoodTV(2023). "MSU researchers get $1.2M to improve 911 call security". April 27.
- EurekAlert (2023). "MSU researchers to lead security improvements for cellular 911 calls with $1.2M NSF grant". April 27.
- MSUToday(2023). "Making emergency calls more secure". April 24.
- HelpNetSecurity (2021). "Bitcoin Security Rectifier app aims to make Bitcoin more secure". May 10.
- StudyFinds (2021). "Bitcoin wallet apps are too vulnerable to use safely, researchers warn". May 5.
- MSUToday(2021). "Making Bitcoin more secure". May 4.
- EurekAlert (2021). "New app makes Bitcoin more secure". May 4.

SERVICE TO THE DEPARTMENT, UNIVERSITY, COMMUNITY, AND PROFESSION

Service to the Department

- Fall 2025 – Present: CS Graduate Admission Committee, Department of Computer Science and Electrical Engineering, UMBC
- Fall 2025: CSEE Lightning Talks and Open House, Department of Computer Science and Electrical Engineering, UMBC

Service to the University

- 2025–present: Member, Asian and Asian American Faculty Staff Council, UMBC

Professional Services

Conference Publicity Chair

- IEEE Conference on Dependable and Secure Computing (DSC) 2025

Conference Workshop Organizer

- AI-Powered Resilient Mobile Health: From AI-RAN to Application, In conjuction with IEEE/ACM CHASE 2026, Pittsburgh, PA, USA. 2025

Conference Technical Program Committee

- IEEE Conference on Communications and Network Security (CNS) 2026
- IEEE Conference on Dependable and Secure Computing (DSC) 2025
- IEEE Conference on Communications and Network Security (CNS) 2025
- IEEE International Conference on Parallel and Distributed Systems (ICPADS) 2025
- First Workshop on Autonomous Drone Computing Systems and Applications (DroneSys, in conjunction with the ACM/IEEE Symposium on Edge Computing) 2025
- IEEE International Joint Conference on Neural Networks (IJCNN) 2024

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC) 2026
- IEEE Transactions on Dependable and Secure Computing (TDSC) 2025
- IEEE Transactions on Information Forensics and Security (T-IFS) 2025
- ACM Journal on Autonomous Transportation Systems (ACM JATS) 2025
- IEEE Transactions on Reliability (IEEE TR) 2025
- ACM Transactions on Internet of Things (ACM TIOT) 2025
- IEEE Transactions on Networking (IEEE TNET) 2025
- IEEE Transactions on Networking (IEEE TNET) 2024
- IEEE Transactions on Reliability (IEEE TR) 2024

- Neural Computing and Applications (NCAA) 2024
- IEEE Transactions on Networking (IEEE TNET) 2023
- IEEE Transactions on Dependable and Secure Computing (TDSC) 2022
- IEEE Transactions on Information Forensics and Security (T-IFS) 2021

Conference Reviewer

- IEEE International Conference on Computer Communications (INFOCOM) 2026
- IEEE Conference on Communications and Network Security (CNS) 2025
- IEEE Conference on Dependable and Secure Computing (DSC) 2025
- IEEE International Conference on Parallel and Distributed Systems (ICPADS) 2025
- IEEE International Joint Conference on Neural Networks (IJCNN) 2024
- IEEE International Conference on Computer Communications (INFOCOM) 2024
- IEEE Military Communications Conference (MILCOM) 2024
- IEEE International Conference on Computer Communications (INFOCOM) 2023
- International Conference on Neural Information Processing (ICONIP) 2023
- IEEE International Conference on Network Protocols (ICNP) 2023
- IEEE Conference on Communications and Network Security (CNS) 2023
- IEEE International Conference on Computer Communications and Networks (ICCCN) 2023
- IFIP Networking Conference (IFIP) 2023
- IEEE International Conference on Communications (ICC) 2023
- IEEE Conference on Big Data (BigData) 2022
- IEEE International Conference on Computer Communications (INFOCOM) 2021
- IEEE Vehicular Networking Conference (VNC) 2020
- IEEE International Conference on Computer Communications (INFOCOM) 2020
- IEEE Global Communications Conference (GLOBECOM) 2019
- IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS) 2019